

JASMINE LEE

Fremont, CA | jas.lee@berkeley.edu | www.linkedin.com/in/jasmineclee/ | jasmineclee.vercel.app/

EDUCATION

University of California, Berkeley

May 2026

B.A. in Data Science, B.A. in Cognitive Science

GPA: 3.9

Berkeley Certificate in *Design Innovation*

Relevant Coursework: Linear Algebra, Probability, Data Engineering, Deep Learning, Industrial and Commercial Data Systems, Cognitive Neuroscience, Human-AI Design Methods, Prototyping and Fabrication

EXPERIENCE

Database Designer

Sept 2025 – Dec 2025

Lawrence Berkeley National Laboratory

Berkeley, CA

- Designed ER diagrams and normalized a relational database (1NF→BCNF) across **25+** tables for large-scale facilities data (i.e. sensors, maintenance logs, controllers).
- Wrote SQL pipelines and queries for failure detection, reliability tracking, and compliance monitoring.
- Structured complex operational data to support scalable asset management and facilities decision-making.

Lead Design Director

Sept 2025 – July 2026

Cal Hacks

Berkeley, CA

- Designed branding and user-facing systems for Cal Hacks' **3,000+** participant flagship hackathon and **1,600+** participant AI Hackathon.
- Led a team of 3 designers and coordinated cross-functional teams to deliver visual and product-facing materials across participant, sponsor, promotional, and live event experiences.

Web Design Lead

Aug 2024 – May 2026

Cognitive Science Students Association

Berkeley, CA

- Designed student-facing planning tools and website redesigns for developer handoff; provided technical support for a **200+** attendee conference.

PROJECTS

GonioMetrix | Jacobs Institute for Design Innovation

2026

- Designed a wearable smart goniometer for independent joint-angle measurement in at-home rehab, athletic training, and ergonomic assessment.
- Translated user research and systems analysis into a prototype with **dual IMU sensing**, ESP32 processing, haptic motors, speaker feedback, and a desktop GUI.
- Implemented a real-time feedback loop that compares movement against target ROM ranges and cues users through non-visual haptic/audio feedback.

Robotic Surgery EEG & Eye-Gaze Modeling | Python, MNE, scikit-learn

2026

- Built an end-to-end multimodal pipeline for robotic-surgery simulator data, preprocessing EEG and eye-gaze from 25 participants for **subject-independent performance prediction**.
- Engineered EEG spectral features using 2-second windows, Welch PSD, band power, and median aggregation, and reduced redundancy via correlation analysis.
- Evaluated gaze, EEG, and multimodal Ridge models with nested leave-one-subject-out cross-validation, isolating participant-level test data from scaling and hyperparameter tuning.

Deep Learning Framework | Python, NumPy, PyTorch, Autograd

2026

- Built a mini deep learning framework from scratch with reverse-mode autograd, computation graphs, tensor operations, modular layers, optimizers, and data loading.
- Implemented core neural network components including Linear, ReLU, BatchNorm, LayerNorm, Dropout, residual blocks, SoftmaxLoss, SGD, and Adam.

TECHNICAL SKILLS

Technical: Python, SQL, Java, Pandas, NumPy, scikit-learn, PyTorch, MongoDB, Git, GitHub

Design: Figma, Adobe Creative Suite (Photoshop, Illustrator, Lightroom), Autodesk Fusion

Interests: Neuropsychology, Wearable Health Systems, Human-AI Interaction, Robotics, Archery